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Computerized Analysis of the Eye Vasculature in a Mass Dataset of Digital Fundus Images: The Example of Age, Sex, and Primary Open-Angle Glaucoma

Jonathan Fhima, MSc,[1,2,*] *Jan Van Eijgen, MD, PhD,*[3,4,*] *Anat Reiner-Benaim, PhD,*[5] *Lennert Beeckmans, MSc,*[3,6] *Or Abramovich, MSc,*[1] *Ingeborg Stalmans, MD, PhD,*[3,4] *Joachim A. Behar, PhD*[1]

Objective: To develop and validate an automated end-to-end methodology for analyzing retinal vasculature in large datasets of digital fundus images (DFIs), aiming to assess the influence of demographic and clinical factors on retinal microvasculature.

Design: This study employs a retrospective cohort design to achieve its objectives.

Participants: The research utilized a substantial dataset consisting of 32 768 DFIs obtained from individuals undergoing routine eye examinations. There was no inclusion of a separate control group in this study.

Methods: The proposed methodology integrates multiple stages: initial image quality assessment, detection of the optic disc (OD), definition of the region of interest surrounding the OD, automated segmentation of retinal arterioles and venules, and the engineering of digital biomarkers representing vasculature characteristics. To analyze the impact of demographic variables (age, sex) and clinical factors (disc size, primary open-angle glaucoma [POAG]), statistical analyses were performed using linear mixed-effects models.

Main Outcome Measures: The primary outcomes measured were changes in the retinal vascular geometry. Special attention was given to evaluating the independent effects of age, sex, disc size, and POAG on the newly engineered microvasculature biomarkers.

Results: The analysis revealed significant independent similarities in the retinal vascular geometry alterations associated with both advanced age and POAG. These findings suggest a potential mechanism of accelerated vascular aging in patients with POAG.

Conclusions: This novel methodology allows for the comprehensive and quantitative analysis of retinal vasculature, facilitating the investigation of its correlations with specific diseases. By enabling the reproducible analysis of extensive datasets, this approach provides valuable insights into the state of retinal vascular health and its broader implications for cardiovascular and ocular health. The software developed through this research will be made publicly available upon publication, offering a critical tool for ongoing and future studies in retinal vasculature.

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The eye is a sensory organ attuned to visible light and a crucial component of the sensory nervous system. Its transparent media serve as a unique noninvasive window to the vascular system, vulnerable to a variety of diseases. The ocular vasculature can be noninvasively examined using digital fundus images (DFIs). This facilitates unprecedented opportunities to analyze the retinal microvasculature and its potential connection to health disorders.

In 1999, Sharrett et al[1] identified arteriovenous nicking and arteriole (A) narrowing as pathological findings of hypertension. Their study involved analysis of the central retinal artery equivalent (CRAE) and the central retinal vein equivalent (CRVE) in a cohort of 9040 patients. In 2006, Witt et al[2] described how the decrease in tortuosity in the retinal As was associated with an increased risk of death due to ischemic heart disease (n = 680) in the Beaver Dam Eye Study. In 2009, Sabanayagam et al[3] reported that individuals with reduced CRAE were more likely to have chronic kidney disease than those with increased CRAE (n = 2581). Regarding primary open-angle glaucoma (POAG), several studies found lower CRAE in patients with POAG compared with healthy controls; however, none of these studies included more than several hundred glaucoma cases.[4–13] Similar to CRAE, but to a lesser extent, lower CRVE is commonly found in this population.[4,6–13] In the subsequent decade, fractal dimension of the retinal vascular tree emerged as a risk

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predictor of cardiovascular disease. Studies conducted between 2008 and 2011 with patient cohorts ranging from 166 to 3303 participants showed that monofractal dimension could vary in association with age, smoking behavior, blood pressure, diabetic retinopathy, chronic kidney disease, stroke, and mortality due to coronary heart disease.[14–19] In 2021, retinal multifractal dimensions (n = 2333) were associated with blood pressure and cardiovascular disease.[20] In patients with POAG, a lower arteriolar and venular fractal dimension was observed compared with healthy controls; however, these studies are small (n = 87 and n = 61 POAG cases).[4,21] In 2022, a study of 1770 patients suggested that a set of vascular biomarkers (VBMs) could undergo changes in response to hypertension. The study calculated 10 VBMs in a semiautomated manner using the SIVA software (Singapore Eye Research Institute).[22] Recent studies have used deep learning to analyze large-scale datasets,[23,24] but these models often lack explainability, impeding physiological insights.

Despite growing advances and the diversity of patient cohorts investigated over the years, these studies share common limitations. The semiautomated nature of the computation might introduce some discrepancies in results depending on the grader and hinder scalability due to the intensive human supervision required for laborious accurate analysis. Furthermore, while CRAE, CRVE, and arteriovenous ratio (AVR) are computed within a specified area of the DFIs, this is not necessarily the case for other biomarkers. Some studies[25,26] have used zones A, B, and C, originally devised for CRAE, CRVE, and AVR measurements, to calculate different VBMs. Nevertheless, this method might be prone to bias as these zones are determined based on the size of the optic disc (OD), which can vary due to conditions like POAG.[27]

This research introduces an end-to-end method for analyzing the retinal vasculature in large DFI datasets. Deep learning techniques are employed to automatically extract the OD and blood vessels from a DFI, as well as to classify these blood vessels into As and venules (Vs). Using these automated extractions, we propose a method for standardizing VBM analysis, based on a fixed-sized region of interest (ROI) around the OD. This minimizes the influence of image centering around the OD. Our methods include statistical analysis using linear mixed models to evaluate the effects of demographics or clinical variables on individual microvascular biomarkers. The method is applied to a dataset of 100 000 DFIs for evaluating the effects of age, sex, and POAG on the microvasculature. This novel approach enables quantitative analysis of ocular vasculature and supports the study of its association with specific diseases, facilitating reproducible analysis of large datasets. The resulting software is made available to the community.

Methods

Study Design and Participants

We used the Leuven dataset[28,29] which consists of a total of 115 237 DFIs from 13 185 unique participants, captured between 2010 and 2019 in the Glaucoma Clinic of the University Hospitals UZ Leuven in Belgium (Institutional Review Board No S60649). The DFIs of the Universitair Ziekenhuis Leuven dataset are disc-centered and acquired using a Visucam Pro NM camera (Zeiss) with a 30° field of view and a resolution of 1444 × 1444 pixels. Low-quality images, defined as FundusQ-Net <6, were excluded.[30] Each of the DFIs was labeled with no, 1, or more medical diagnosis codes (International Classification of Diseases 10). The codes were reviewed and categorized as healthy, POAG, or other as defined in section Study Design and Participants. Digital fundus images were excluded if the OD or A/V could not be segmented, or if the ROI was <500 pixels in radius, primarily due to image centering issues that omitted the OD and were largely caused by acquisition errors unrelated to disease. These excluded images represent <4% of the dataset, and their small size minimizes any potential impact. This led to a final dataset containing 32 768 DFIs from 4858 unique participants. Figure 1 summarizes the data selection steps.

Children <18 years of age and patients with diabetic or hypertensive retinopathy were excluded from the dataset. Primary open-angle glaucoma–labeled eyes, either normal-tension glaucoma or high-tension glaucoma based on the maximal untreated intraocular pressure (21 mmHg cutoff), were included until or unless they were diagnosed with other eye diseases, congenital diseases, nonglaucoma surgery, or interventions. The absence of disease labels resulted in the classification of the healthy group.

Automated Biomarkers Extraction

LUNet[28] was used to segment the OD and A/V from the DFI. An ROI with a fixed size of 500 pixels was extracted around the OD (excluding the OD), along with zones A and B, for all DFIs. Subsequently, the CRAE, CRVE, and AVR were calculated using the A/V from zone B. To correct for variability of the zones A and B, the radius of zone A, named disc radius A, was adjusted for. In contrast, other VBMs were computed within the fixed-size ROI for the As and Vs (Fig S2, available at www.ophthalmologyscience.org). The overall pipeline is summarized in Figure 3.

To define a large-scale vasculature analysis pipeline, it is necessary to segment the A/V and the OD from the DFIs. LUNet[28] was used to perform a high-resolution A/V segmentation. LUNet was trained to perform A/V segmentation at the same resolution as the DFIs from the Universitair Ziekenhuis Leuven dataset. Briefly, LUNet architecture includes a double dilated convolutional block that aims to enhance the receptive field of the model and reduce its parameter count.

The custom loss of function emphasizes the continuity of blood vessels. Optic disc segmentation does not require the same fine-grained segmentation as the A/V task. For efficient segmentation of the OD, LUNet was trained on DFIs that were downscaled to a resolution of 256 × 256 pixels. This downsizing of the DFIs was carried out using bilinear interpolation. Subsequently, the probability map generated by LUNet was upscaled back to the original resolution of the DFIs using bilinear interpolation, resulting in a segmentation map that matched the original dimensions of the DFIs. An example of DFI segmentation can be seen in Figure S2.

In POAG follow-up, DFIs are often disc-centered. However, this centering is never exact, and this affects the estimation of several VBMs. To standardize the biomarker computation and minimize this effect, an ROI centered on the OD is defined. The ROI radius was set to 500 pixels, which resulted in 95% retention of the segmentation for analysis (see Fig S4, available at www.ophthalmologyscience.org). An example of the ROI extracted is presented in Figure 5. Finally, the VBM was engineered for A/V separately using the Python Vasculature BioMarker[31] toolbox. The set of VBM included is summarized

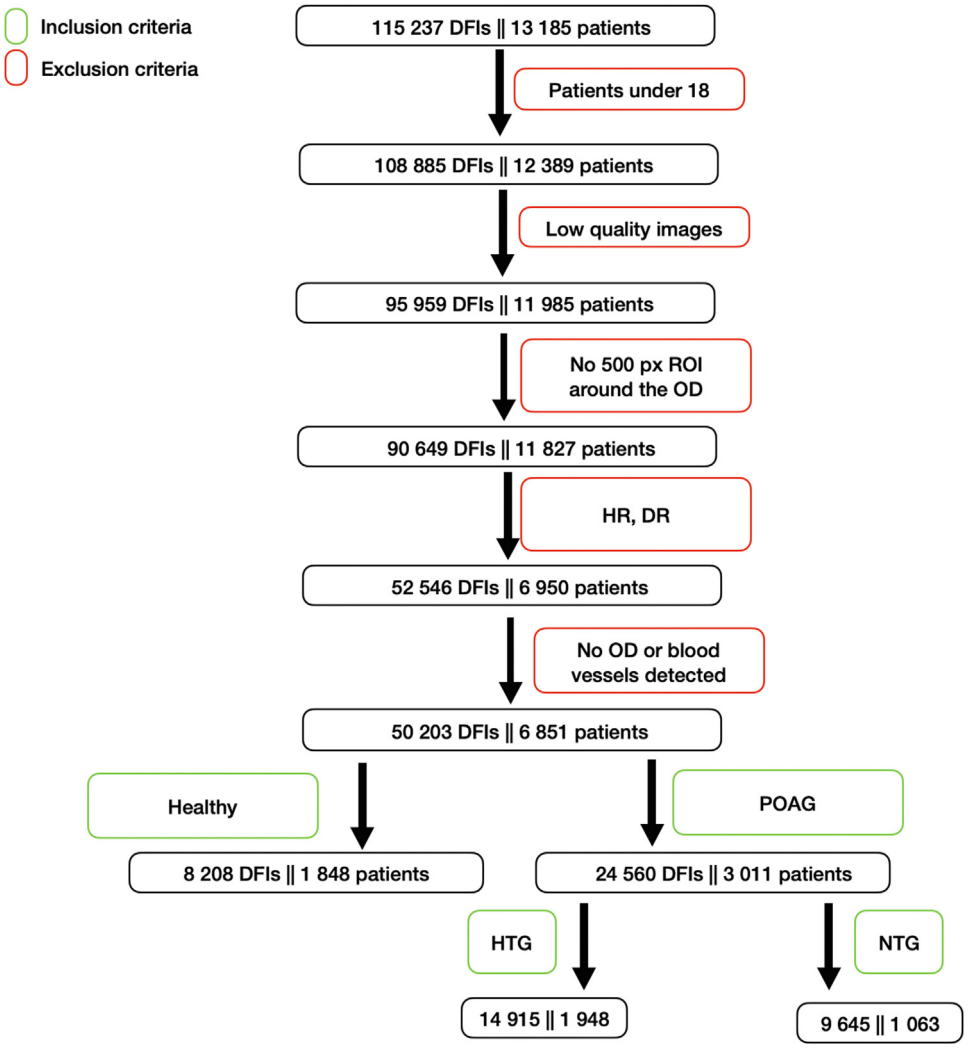


Figure 1. Summary of the dataset elaboration. DFI = digital fundus image; DR = diabetic retinopathy; HR = hypertensive retinopathy; HTG = high-tension glaucoma; NTG = normal-tension glaucoma; OD = optic disc; ROI = region of interest.

in Table 1. Examples of engineered VBMs inside the ROI can be seen in Figure 3.

Statistical Analysis

Digital VBM levels were compared between healthy and POAG eyes using the Mann−Whitney *U* test as all biomarkers were nonnormally distributed (Anderson−Darling test). Since the groups differed in age and sex, a multivariate linear mixed-effects model (LMM) was used to evaluate and correct for the effect of age, sex, disc radius, and POAG on each VBM. The random effects of the LMM additionally account for the dependence between multiple DFIs of the same patient. Based on the central limit theorem, LMM analysis can be validly interpreted. The LMM was trained on the complete dataset of 32 768 images. The resulting standardized model coefficients (Z statistics) were presented for all VBMs as a heatmap. For the Mann−Whitney *U* tests, only the first image of every patient was used, randomly sampled for the left or right eye if both were eligible for analysis. The *P* values were adjusted for multiplicity using the Benjamini−Hochberg false discovery rate controlling procedure.[39] The 0.05 level was used for significance.

Results

After selection, 3010 patients with POAG and 1848 healthy controls were included in this experiment. The average age of the POAG cohort (70 years) was higher than that of the control cohort (59 years, $P < 0.001$), as was the proportion of men (47% vs. 40% respectively, $P < 0.001$). The demographics, as well as the proportion of normal-tension glaucoma and high-tension glaucoma, can be found in Table 2. Overall, the POAG cohort is characterized by moderate disease. As age, sex, and disc radius differ between groups, adjustment for these covariates is necessitated.

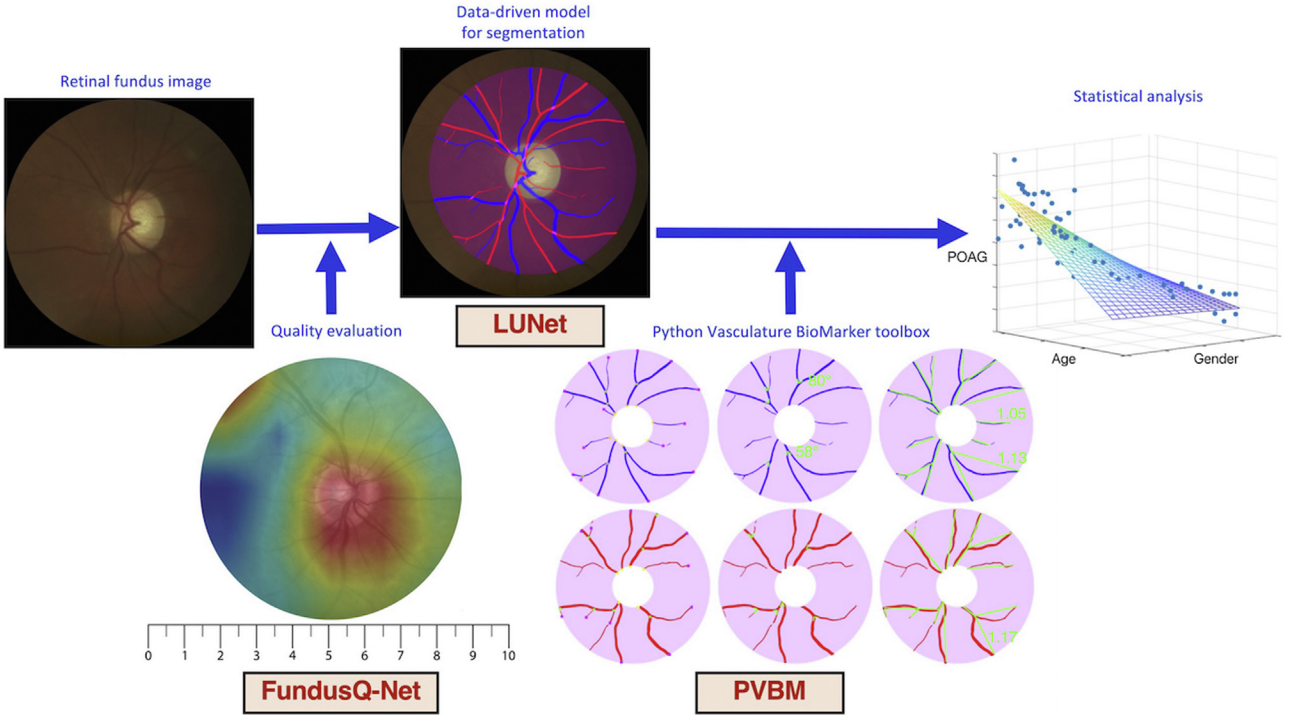


Figure 3. Overview of the experiments. The quality of the digital fundus images is first automatically computed using FundusQ-Net.[30] Digital fundus images of sufficient quality are processed with LUNet[29] for arteriole/venule and optic disc segmentation. A region of interest is defined around the optic disc to standardize the analysis, and a set of digital vasculature biomarkers is computed using the PVBM toolbox.[31] Finally, a statistical analysis of the vasculature characteristics is performed as a function of demographic and clinical variables. POAG = primary open-angle glaucoma; PVBM = Python Vasculature BioMarker.

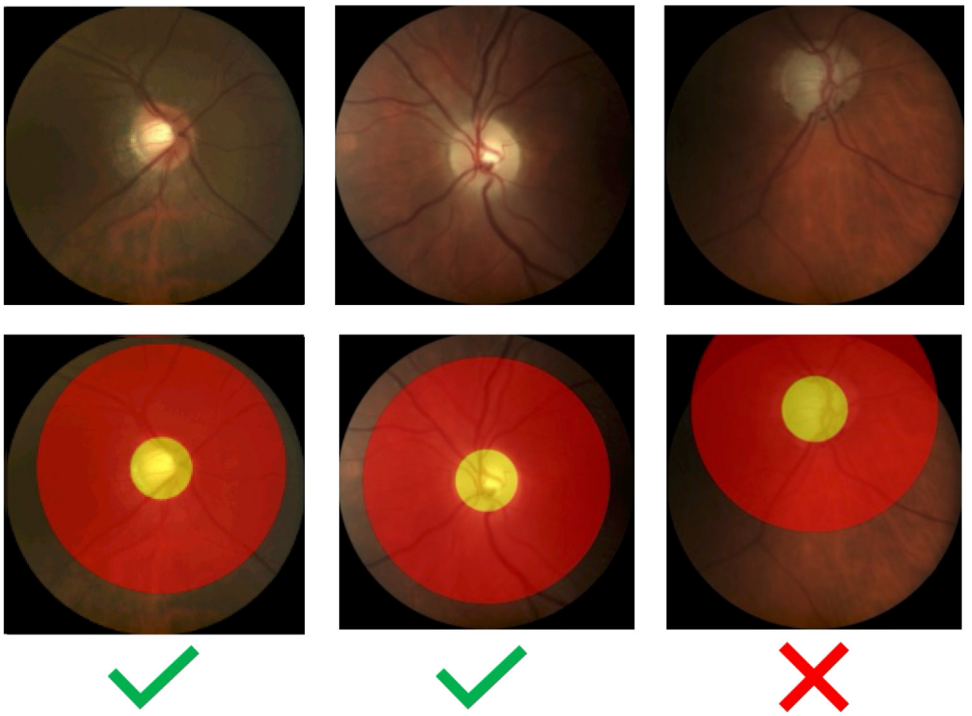


Figure 5. Example of DFIs accepted and discarded based on the availability of the ROI. DFI = digital fundus image; ROI = region of interest.

Table 1. List of Digital Vasculature Biomarkers Implemented in PVBM[31]

Number	Biomarker	Definition	Unit
1	OVLEN[32]	Overall length	Pixel
2	AREA[32]	Overall area	Pixel^2
3	START[32]	Number of starting points	-
4	END[32]	Number of end points	-
5	INTER[32]	Number of intersection points	-
6	TOR[32,33]	Median tortuosity	%
7	TI[32]	Tortuosity index	%
8	BA[32]	Branching angles	*Degree*
9	D_0[34,35]	Capacity dimension	-
10	D_1[34,35]	Entropy dimension	-
11	D_2[34,35]	Correlation dimension	-
12	SL[36]	Singularity length	-
13	$CRAE_h$[37]	Central retinal artery equivalent using the formulas defined by Hubbard et al.	Pixel
14	$CRVE_h$[37]	Central retinal artery equivalent using the formulas defined by Hubbard et al.	Pixel
15	AVR_h[37]	Arteriovenous ratio using the formulas defined by Hubbard et al.	-
16	$CRAE_k$[38]	Central retinal arterial equivalent using the formulas defined by Knudtson et al.	Pixel
17	$CRVE_k$[38]	Central retinal venular equivalent using the formulas defined by Knudtson et al.	Pixel
18	AVR_k[38]	Arteriovenous ratio using the formulas defined by Knudtson et al.	-
19	AVR_h[38]	Arteriovenous ratio using the formulas defined by Hubbard et al.	-

PVBM = Python Vasculature BioMarker.

Table 3 provides the summary statistics for the VBMs of the eyes of patients with POAG and healthy participants with a comparison between the groups. All VBMs are lower in patients with POAG compared with healthy controls, except for arteriolar and venular singularity length and the Knudtson AVR, after adjusting for multiple testing. The multivariate linear mixed models equally show that the VBMs are influenced by POAG, age, sex, and the disc radius A. These results are presented in Figure 6. With age all VBMs are uniformly lower except for singularity length and branching angle. When POAG status, age, and disc diagnosis are held constant, men have lower arteriolar and venular diameter, area and length, lower arteriolar and venular monofractal and multifractal dimensions, lower venular branching angles, and higher arteriolar branching angles. Larger disc size does significantly impact the VBMs, resulting in more starting points and less overall vessel area, which prompts proper statistical adjustment. Although the arteriolar and venular area and length are lower, the CRAE and CRVE tend to be slightly larger.

A boxplot illustrating the arteriolar area, venular tortuosity, CRAE, and CRVE (according to Knudtson method) segmented by diagnosis and sex relative to patient age is depicted in Figure 7. Additional boxplots covering other VBMs are available in the Appendix, as shown in Figure S8 (available at www.ophthalmologyscience.org). These boxplots show a trend in the VBMs with respect to the age, sex, and POAG diagnosis, illustrating the outcome of the statistical analysis.

Discussion

In this work, we propose an end-to-end pipeline for the automated analysis of the retinal vascular geometry in a large population cohort, for which POAG was used as a case example. Full automatization differs from previous work in which the VBM computation was semisupervised.[1–3,9,14–19,40]

All computed retinal vascular parameters are lower in participants of higher age, concordant with preexisting literature.[14,21,41,42] Women have slightly higher arteriolar and venular areas ($Area_a$), due to longer and wider vessels ($OVLEN_a$), and slightly higher arteriolar and venular fractal dimension levels. To our knowledge, this is the

Table 2. Demographics by Diagnosis

Demographic Variable	Excluded (N = 82 569)	Normal (N = 1848)	POAG (N = 3010)	POAG Subtypes	
				NTG (N = 1063)	*HTG (N = 1948)*
Age (yrs, mean ± std)	58.06 ± 21.08	58.54 ± 16.92	69.67 ± 11.39	70.13 ± 11.34	69.42 ± 11.42
Sex (M) MD (dB, mean ± std)	49.6%	38.7%	47.7%	38.0%	53.9%
	-	-	L: −9.01 ± 7.76	L: −8.36 ± 7.06	L: −9.39 ± 8.12
			R: −8.38 ± 7.73	R: −7.68 ± 7.06	R: −8.78 ± 8.06

dB = decibel; HTG = high-tension glaucoma; L = left eye; M = male; MD = mean deviation; NTG = normal-tension glaucoma; POAG = primary open-angle glaucoma; R = right eye; std = standard deviation.

Table 3. Summary of the Computed VBMs Using Their Implementation from PVBM

Biomarker		POAG (n = 3010)		Healthy (n = 1848)		P Values
		Mean	*Std*	*Mean*	*Std*	MWU Test (FDR)
Arterioles	Area	40 231.26	6112.93	45 555.36	6817.43	<0.001
	TI	1.0856	0.0133	1.0890	0.0149	<0.001
	TOR	1.0795	0.0077	1.0814	0.0084	<0.001
	OVLEN	3532.90	597.35	3712.54	618.40	<0.001
	BA	77.55	10.31	78.88	10.31	<0.001
	START	7.44	1.53	7.70	1.49	<0.001
	END	12.38	2.73	13.53	3.23	<0.001
	INTER	5.28	2.59	6.48	3.37	<0.001
	D_0	1.24	0.0345	1.25	0.0350	<0.001
	D_1	1.22	0.0352	1.24	0.0347	<0.001
	D_2	1.22	0.0354	1.23	0.0346	<0.001
	SL	0.886	0.3106	0.860	0.2802	0.371
	CRAE (Hubbard)	29.32	4.44	32.72	4.75	<0.001
	CRAE (Knudtson)	25.94	3.73	28.88	4.03	<0.001
Venules	Area	32 902.50	6401.33	38 247.43	6785.99	<0.001
	TI	1.0848	0.0195	1.0933	0.0230	<0.001
	TOR	1.0789	0.0109	1.0830	0.0128	<0.001
	OVLEN	3496.39	688.88	3701.66	694.32	<0.001
	BA	74.75	11.97	77.80	12.37	<0.001
	START	7.47	1.67	7.76	1.67	<0.001
	END	11.60	2.59	12.47	2.75	<0.001
	INTER	4.42	2.25	5.19	2.59	<0.001
	D_0	1.23	0.0386	1.25	0.0383	<0.001
	D_1	1.22	0.0404	1.24	0.0397	<0.001
	D_2	1.22	0.0411	1.23	0.0402	<0.001
	SL	0.907	0.3107	0.868	0.2704	0.097
	CRVE (Hubbard)	43.86	9.41	45.33	8.77	<0.001
	CRVE (Knudtson)	23.22	3.66	25.95	3.58	<0.001
	AVR (Knudtson)	1.13	0.22	1.12	0.16	0.080
	AVR (Hubbard)	0.73	0.33	0.76	0.24	<0.001
	Sex	0.533	0.499	0.595	0.491	<0.001
	Age	69.66	11.32	58.60	16.91	<0.001

AVR = arteriovenous ratio; BA = branching angles; CRAE = central retinal artery equivalent, CRVE = central retinal vein equivalent; D0, capacity dimension; D1 = entropy dimension; D2 = correlation dimension; END = number of end points; FDR = false discovery rate; INTER = number of intersection points; MWU = Mann–Whitney *U* test; OVLEN = overall length; POAG = primary open-angle glaucoma; PVBM = Python Vasculature BioMarker; SL = singularity length; START = number of starting points; Std = standard deviation; TI = tortuosity index; TOR = median tortuosity. *P* values are FDR adjusted for comparison between POAG and healthy controls using the Mann–Whitney *U* test.

first time significant sex differences have been reported for these biomarkers.

The Beaver Dam Eye study[42] found narrower CRAE and CRVE in men and the Multi-Ethnic Study of Atherosclerosis[41] found narrower CRAE in men, which we replicated. Incident cardiovascular comorbidity (e.g., hypertension prevalence) between the sexes is not reported in these studies, as in the current work, and could represent a potential bias.

The work by Zhu et al[43] is a recent example of the fact that age can be estimated from DFIs using a deep learning model (correlation of 0.81, $P < 0.001$). Similarly, previous work showed that sex can be identified from DFI.[44] For these tasks, end-to-end deep learning models are used and the gain in performance is reached at the expense of explainability. The automated approach presented in this work at least partly tackles the explainability limitation of end-to-end deep learning models, e.g., age and sex estimation characterizing age- and sex-related retinal vascular changes.

Despite extensive research efforts,[45,46] the precise mechanism underlying POAG remains elusive. The mechanical theory posits that mechanical stress induces damage to the ganglion cells and their axons. Factors such as corneal thickness, corneal hysteresis, and the pressure gradient across the lamina cribrosa contribute to the susceptibility of the optic nerve to fluctuations in intraocular pressure, resulting in axonal injury along the optic nerve.[47–49] Alternatively, possibly complimentary, the vascular theorem suggests that optic neuropathy arises from repetitive ischemic insults or subtle changes in perfusion.[50–52] The independent association between elevated arterial blood pressure variability and extremes of blood pressure levels, and the incidence and progression of POAG, alongside the elevated prevalence of cardiovascular or renovascular or neurovascular disease (hyperlipidemia,

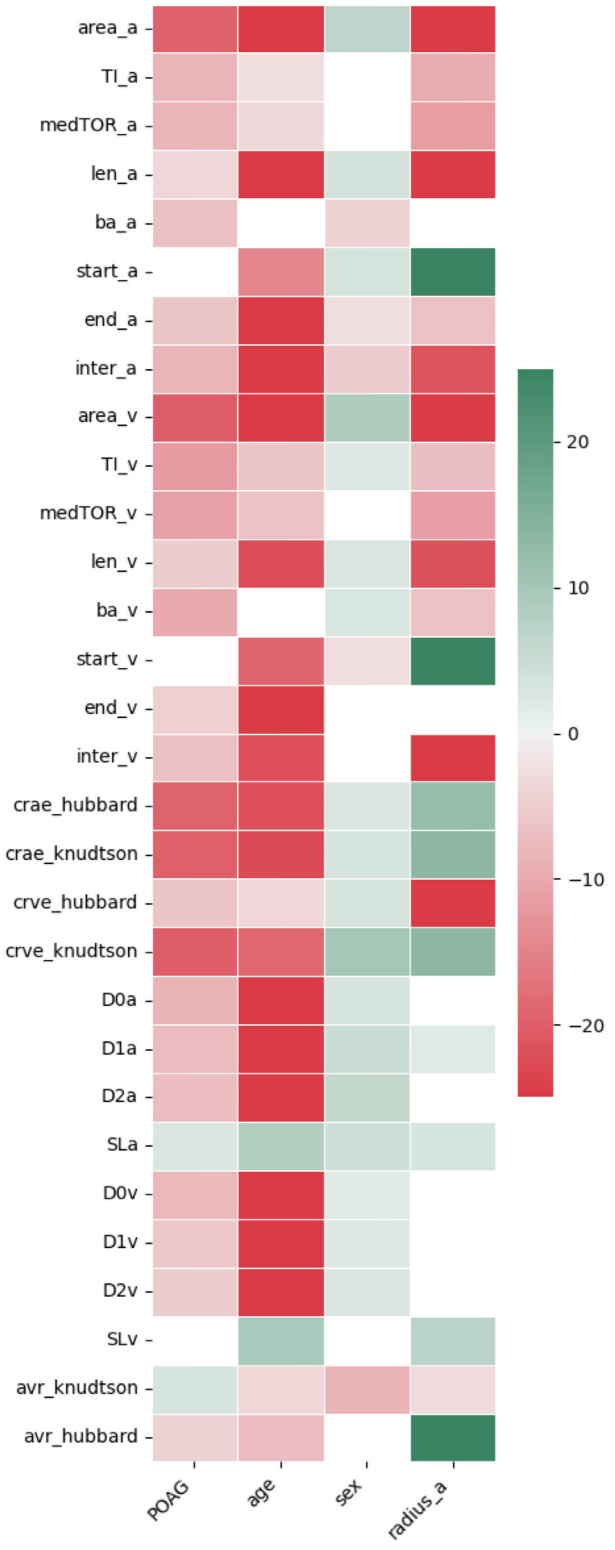


Figure 6. Results of the statistical analysis. Nonsignificant *P* values are in white. Significant *P* values are represented by Z values in the heatmap. Z values >25 or <−25 are clipped to these color values. The features include geometric and fractal descriptors: overall length (OVLEN), overall area (AREA), number of starting points (START), end points (END), and intersections (INTER); vessel tortuosity (TOR) and tortuosity index (TI); branching angles (BA); fractal dimensions (capacity dimension D0, entropy dimension D1, and correlation dimension D2); singularity length (SL); and vascular caliber metrics: central retinal artery equivalent (CRAE) and central retinal vein equivalent (CRVE) calculated according to Hubbard et al (CRAEh, CRVEh) and Knudtson et al (CRAEk, CRVEk), with corresponding arteriovenous ratios (AVRh; AVRk). Indexes _a and _v referring to arterioles and venules, respectively. CRAE = central retinal artery equivalent; CRVE = central retinal vein equivalent; POAG = primary open-angle glaucoma.

metabolic syndrome, ischemic heart disease, stroke small vessel disease, chronic kidney disease) in patients with POAG, provide further support for the interrelation between POAG and systemic vascular dysregulation.[53–56] This work is the first to explore the retinal vascular features of patients with glaucoma on a large scale, thus facilitating a more robust interpretation and statistical validity of vascular theory.

Several studies indicated significantly narrower CRAE and CRVE in patients with POAG compared with controls without POAG.[4,6,10–13,57] Furthermore, a decrease in CRAE was previously associated with an increased incidence of POAG after 10 years of follow-up.[5,9] In patients with asymmetric disease, CRAE and CRVE were narrower in the eye with more severe disease.[8] In bilateral glaucoma suspects who converted to unilateral glaucoma, a narrower CRAE is found at baseline and both narrower CRAE and CRVE are found at the time of conversion, compared with nonconverted eyes. Central retinal artery equivalent and CRVE were significantly narrower at the time of conversion than at baseline, at least in the converted eyes.[7] Yoo et al[12] showed that CRAE had the largest area under the receiver-operating characteristic curve compared with CRVE and AVR for discriminating between patients with POAG and healthy eyes, showing no significant difference with area under the receiver-operating characteristic curve of retinal nerve fiber layer (RNFL) thickness.[12] On the other hand, several studies, including the Rotterdam Study and the Beaver Dam Eye Study, report an insignificant association between CRAE/CRVE and POAG.[42,58–60] Furthermore, several more studies, including the Blue Mountains Eye study, report a nonsignificant relationship between CRVE and POAG prevalence.[4,5,9,11,12] Of note, all these studies consistently had relatively low sample sizes—mostly <100 POAG cases and never surpassing >250 cases—and exhibited clear heterogeneity in vessel segmentation method and the quality of retinal images. This work, comparing multiple images of >3000 patients with POAG and >1800 controls, with a high level of standardization and quality control, can be seen as a definitive conclusion that both CRAE and CRVE are, on average, lower in patients with POAG after adjustment for age, sex, and disc radius A. Consequently, AVR did not significantly differ between patients with POAG and controls, in line with a lowering of both CRAE and CRVE.

Tortuosity, branching angle, and fractal dimension reflect the circulatory optimality of the retinal microvasculature, optimizing blood flow with minimal energy expenditure.[15,61] Tortuosity is associated with tissue hypoxia, potentially mediated by secretions from vascular

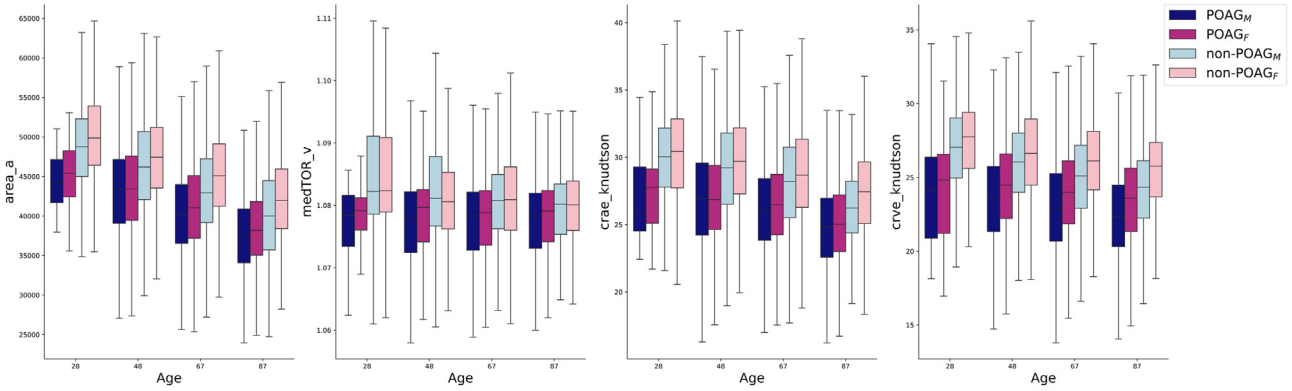


Figure 7. Boxplot of the $Area_a$, TOR_v, CRAE, and CRVE (Knudtson), grouped by diagnosis and gender with respect to the patient age. CRAE = central retinal artery equivalent. CRVE = central retinal vein equivalent; F = female; M = male; POAG = primary open-angle glaucoma; TORV = median tortuosity of the venules.

endothelial cells, which regulate blood flow through mediators such as nitric oxide and endothelin, which influence angiogenesis and therefore, vessel geometry.[62–64] Lower tortuosity has been associated with aging.[65] Some studies, including the European Prosepective Investigation of Cancer-Norfolk Eye Study, did not show a significant association between arteriolar and venular tortuosity in patients with POAG compared with controls.[11,13,59] However, straighter and less tortuous As and Vs were significantly associated with POAG in other trials and are confirmed in this work.[21,58] In the literature, the branching angle was found to be significantly different on the arteriolar or venular side in patients with POAG compared with healthy controls.[21,58] Increased branching angles have been related to a decreased blood flow, whereas decreased angles have been related to aging and hypertension.[66,67] This study confirms both the lower arteriolar and venular branching angles in patients with POAG. The lower fractal dimension, both arteriolar and venular, as measured with OCT angiography, was previously observed in patients with POAG compared with controls and is confirmed in this work.[13,21,68]

In conclusion, after adjustment for age, sex, and disc radius A, patients with POAG seem to exhibit lower arteriolar and venular diameter, area, tortuosity, branching angle, endpoints, intersection points, and fractal dimension levels. These changes mimic those seen in higher age categories, which underline the possibility of pronounced vascular aging and dysfunctional vascular endothelium in patients with POAG.

Limitations and Future Directions

The vascular effects shown in this paper do not prove the vascular theorem but point to a certain association. Similarly, Tham et al[10] observed no association between CRAE and POAG prevalence after adjusting for RNFL thickness, in contrast with CRVE, which showed a significant association after adjustment. For both CRAE and CRVE, the indirect effects, being the associations mediated through RNFL thickness, were significant and were 32.3% and 12.3% of the total effect, respectively. In analogy, this association cannot necessarily be used in favor of the vascular theorem as the vascular changes can occur secondary to preceding RNFL loss due to other reasons. Longitudinal analyses of retinal vascular changes and RNFL thinning in glaucoma progression have the potential to further elucidate this discussion and represent the subject of future research. The disc size did influence the parameters. As the disc radius increased, the vessel area and length, calculated in a fixed ROI excluding the OD, logically decreased and the number of starting points increased. Both CRAE and CRVE, calculated in zone B, which depends on the disc radius A, tended to increase in higher radii. These differences point to the need for adjustment of the disc size in automated VBM pipelines.

This is the first time a fully automated end-to-end pipeline is published for the analysis of retinal vascular geometry in a large cohort (13 185 patients). Both arteriolar and venular diameter, area, length, tortuosity, branching angle, end points, intersection points, and fractal dimension levels are independently lower in patients with POAG and older patients, after adjustment for sex and disc radius.

Given the independent similarities in retinal vascular geometry changes between older age and POAG prevalence, the theory of pronounced vascular aging in patients with POAG is proposed. Further exploration of vascular changes across different POAG subgroups represents an important avenue for future research.

Despite age and sex adjustments, demographic imbalance between control and glaucoma groups may still influence the findings. This limitation should be considered when interpreting the results. Additionally, the accuracy and generalizability of the estimation of the vasculature biomarkers depends on the performance of the vascular segmentation model in extracting the As and the Vs from a fundus image. An important limitation in our work is the lack of systematic human expert validation to ensure the

quality of the automated segmentations. Additionally, the current version of LUNet was trained to work on DFI with a field of view of 30°. Other fields of view (e.g., 15°, 45°) are used in clinical practice and the performance of LUNet may be suboptimal on such images. This limitation of LUNet will be addressed in future work.

Although all images were captured using the same camera model and standardized protocol, factors like axial length, keratometry, and camera-to-eye distance may have affected the resolution. While detailed imaging parameters were unavailable, the consistent imaging setup minimizes variability, enabling reliable relative comparisons within the dataset.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the authors used ChatGPT, an AI-based language model developed by OpenAI, in order to edit the English language of this manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Footnotes and Disclosures

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[1] Faculty of Biomedical Engineering, Technion-IIT, Haifa, Israel.

[2] Department of Applied Mathematics, Technion-IIT, Haifa, Israel.

[3] Research Group Ophthalmology, Department of Neurosciences, KU Leuven, Leuven, Belgium.

[4] Department of Ophthalmology, University Hospitals UZ Leuven, Leuven, Belgium.

[5] Department of Epidemiology, Biostatistics, and Community Health Sciences, School of Public Health, Faculty of Health Sciences, Ben-Gurion University of the Negev, Beer-Sheva, Israel.

[6] Department of Electrical Engineering (ESAT), KU Leuven, Leuven, Belgium.

*J.F. and J.V.E. contributed equally to this work.

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No animal subjects were included in this study.

Author Contributions:

Conception and design: Fhima, Eijgen, Reiner-Benaim, Beeckmans, Abramovich, Stalmans, Behar

Data collection: Fhima, Eijgen, Reiner-Benaim, Beeckmans, Abramovich, Stalmans, Behar

Analysis and interpretation: Fhima, Eijgen, Reiner-Benaim, Beeckmans, Abramovich, Stalmans, Behar

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Overall responsibility: Fhima, Eijgen, Reiner-Benaim, Beeckmans, Abramovich, Stalmans, Behar

Abbreviations and Acronyms:
A = arteriole; **AVR** = arteriovenous ratio; **CRAE** = central retinal artery equivalent; **CRVE** = central retinal vein equivalent; **DFI** = digital fundus images; **LMM** = linear mixed-effects model; **OD** = optic disc; **POAG** = primary open-angle glaucoma; **RNFL** = retinal nerve fiber layer; **ROI** = region of interest; **V** = venule; **VBM** = vascular biomarker.

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Correspondence:
Joachim A. Behar, PhD, Faculty of Biomedical Engineering, Julius Silver Building, Floor 1, Office 121, Technion - IIT, Haifa 3200003, Israel. E-mail: jbehar@technion.ac.il.

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